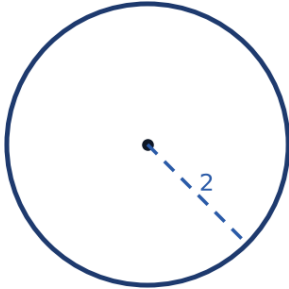


Polar Curves and Derivatives



Converting between polar and rectangular coordinates

- 1 Convert the polar point $(4, \frac{\pi}{3})$ to rectangular coordinates.
- 2 Convert the polar equation $r = 4\cos \theta$ to rectangular form, and identify the curve.



Identifying and sketching common polar curves

- 3 Identify the type of curve for $r = 2 + 2\sin \theta$, and state its key feature.
- 4 Identify the type of curve for $r = \cos(3\theta)$.

The slope of a polar curve

- 5 State the formula for $\frac{dy}{dx}$ of a polar curve $r = f(\theta)$, using $x = r\cos \theta$ and $y = r\sin \theta$.
- 6 Find $\frac{dy}{dx}$ for $r = 2\theta$ at $\theta = \frac{\pi}{2}$.

Finding horizontal and vertical tangents

- 7 For the circle $r = 4\cos \theta$, find the value(s) of θ in $[0, \pi]$ where the tangent line is horizontal (where $y = r\sin \theta$ has a critical point).

Symmetry of polar curves

- 8 Explain how to test whether a polar curve $r = f(\theta)$ is symmetric about the polar axis (the x -axis).
- 9 Test whether $r = 3\sin \theta$ is symmetric about the y -axis (replace θ with $\pi - \theta$).
-

More polar-to-rectangular conversions

- 10 Convert the rectangular point $(3, 3\sqrt{3})$ to polar coordinates ($r > 0, 0 \leq \theta < 2\pi$).
- 11 Convert the rectangular equation $y = x$ to polar form.
- 12 Convert the polar equation $r\sin \theta = 4$ to rectangular form, and identify the curve.
-

More slope-of-a-polar-curve practice

- 13 Find $\frac{dy}{dx}$ for the cardioid $r = 1 + \sin \theta$ at $\theta = 0$.
- 14 Find $\frac{dy}{dx}$ for the circle $r = 5$ (constant) at any angle θ , and interpret the result using the fact that a circle centered at the origin has radial symmetry.
-

Tangent lines at the pole

- 15 Find the values of θ where the cardioid $r = 1 - \cos \theta$ passes through the pole ($r = 0$), and explain why these give the tangent line directions at the pole.
- 16 Find the values of $\theta \in [0, 2\pi)$ where the rose $r = \cos(2\theta)$ passes through the pole.
-

Classifying polar curves by equation type

- 17 Classify each polar equation by curve type (circle, cardioid, limaçon with inner loop, rose, or line):
- | | |
|--------------------------|--------------------------|
| a $r = 5$ | b $r = 3 + 3\cos \theta$ |
| c $r = 1 + 3\sin \theta$ | d $r = 4\sin(5\theta)$ |
-

Number of petals in rose curves

- 18 State the rule for the number of petals of $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$ depending on whether n is odd or even.

- 19 How many petals does $r = 2\sin(4\theta)$ have? How many does $r = 5\cos(7\theta)$ have?
-

Graphing polar curves point-by-point

- 20 Complete a short table of (θ, r) values for $r = 2 + 2\cos\theta$ at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, and use it to describe the cardioid's shape.
-

Concavity considerations for polar curves

- 21 Explain, in terms of $\frac{dy}{dx}$, what a 'vertical tangent' means for a polar curve, and why it corresponds to $\frac{dx}{d\theta} = 0$ (with $\frac{dy}{d\theta} \neq 0$).
-

Multiple choice — easy 15

- 22 How many petals does the rose $r = \cos(2\theta)$ have?
- a. 4
 - b. 6
 - c. 7
 - d. 2
- 23 How many petals does the rose $r = 7\cos(6\theta)$ have?
- a. 14
 - b. 10
 - c. 6
 - d. 12

24 How many petals does the rose $r = 4\cos(7\theta)$ have?

a. 14

b. 9

c. 7

d. 5

25 How many petals does the rose $r = 4\cos(2\theta)$ have?

a. 7

b. 2

c. 4

d. 6

26 How many petals does the rose $r = 6\cos(2\theta)$ have?

a. 7

b. 4

c. 2

d. 6

27 How many petals does the rose $r = 3\sin(7\theta)$ have?

a. 9

b. 7

c. 14

d. 5

28 How many petals does the rose $r = 6\sin(7\theta)$ have?

- a. 14
- b. 5
- c. 7
- d. 9

29 How many petals does the rose $r = 9\cos(4\theta)$ have?

- a. 8
- b. 6
- c. 4
- d. 10

30 How many petals does the rose $r = 3\sin(4\theta)$ have?

- a. 6
- b. 4
- c. 10
- d. 8

31 How many petals does the rose $r = 2\sin(3\theta)$ have?

- a. 6
- b. 5
- c. 1
- d. 3

32 How many petals does the rose $r = 5\cos(2\theta)$ have?

a. 6

b. 4

c. 1

d. 2

33 How many petals does the rose $r = 2\cos(4\theta)$ have?

a. 6

b. 4

c. 10

d. 8

34 How many petals does the rose $r = 2\cos(8\theta)$ have?

a. 14

b. 18

c. 16

d. 8

35 How many petals does the rose $r = 2\sin(6\theta)$ have?

a. 14

b. 12

c. 10

d. 6

36 How many petals does the rose $r = 7\cos(7\theta)$ have?

- a. 5
- b. 9
- c. 7
- d. 14

Multiple choice — medium 15

37 Convert the polar point $(-8, \frac{3\pi}{2})$ to rectangular coordinates.

- a. $(-1, 10)$
- b. $(8, 0)$
- c. $(0, 8)$
- d. $(0, -8)$

38 Convert the polar point $(-2, \frac{3\pi}{2})$ to rectangular coordinates.

- a. $(0, 2)$
- b. $(2, 0)$
- c. $(-2, 3)$
- d. $(0, -2)$

39 Convert the polar point $(6, \pi)$ to rectangular coordinates.

- a. $(0, -6)$
- b. $(6, 0)$
- c. $(-6, 0)$
- d. $(-5, 0)$

40 Convert the polar point $(2, \frac{\pi}{2})$ to rectangular coordinates.

- a. $(2, 0)$
- b. $(0, 2)$
- c. $(0, -2)$
- d. $(2, 4)$

41 Convert the polar point $(-8, \pi)$ to rectangular coordinates.

- a. $(9, 0)$
- b. $(0, 8)$
- c. $(-8, 0)$
- d. $(8, 0)$

42 Convert the polar point $(-5, \frac{3\pi}{2})$ to rectangular coordinates.

- a. $(5, 0)$
- b. $(0, 5)$
- c. $(0, -5)$
- d. $(2, 3)$

43 Convert the polar point $(7, \pi)$ to rectangular coordinates.

- a. $(7, 0)$
- b. $(0, -7)$
- c. $(-6, 0)$
- d. $(-7, 0)$

- 44 Convert the polar point $(2, \pi)$ to rectangular coordinates.
- a. $(2, 0)$
 - b. $(-2, 0)$
 - c. $(-1, 0)$
 - d. $(0, -2)$
- 45 Convert the polar point $(4, 0)$ to rectangular coordinates.
- a. $(0, 4)$
 - b. $(5, 0)$
 - c. $(-4, 0)$
 - d. $(4, 0)$
- 46 Convert the polar point $(5, 0)$ to rectangular coordinates.
- a. $(6, 0)$
 - b. $(-5, 0)$
 - c. $(5, 0)$
 - d. $(0, 5)$
- 47 Convert the polar point $(-3, 0)$ to rectangular coordinates.
- a. $(-3, 0)$
 - b. $(3, 0)$
 - c. $(-2, 0)$
 - d. $(0, -3)$

48 Convert the polar point $(4, \frac{\pi}{2})$ to rectangular coordinates.

- a. $(0, -4)$
- b. $(1, 3)$
- c. $(0, 4)$
- d. $(4, 0)$

49 Convert the polar point $(9, \frac{\pi}{2})$ to rectangular coordinates.

- a. $(0, 9)$
- b. $(0, -9)$
- c. $(9, 0)$
- d. $(2, 11)$

50 Convert the polar point $(-6, \pi)$ to rectangular coordinates.

- a. $(0, 6)$
- b. $(-6, 0)$
- c. $(7, 0)$
- d. $(6, 0)$

51 Convert the polar point $(-5, \frac{\pi}{2})$ to rectangular coordinates.

- a. $(-5, 0)$
- b. $(0, -5)$
- c. $(0, 5)$
- d. $(2, -3)$

Multiple choice — hard 15

52 Classify the curve $r = \sin \theta$.

- a.** Cardioid
- b.** Limaçon without an inner loop
- c.** Circle through the pole
- d.** Limaçon with an inner loop

53 Classify the curve $r = 9 + 9\cos \theta$.

- a.** Circle through the pole
- b.** Limaçon with an inner loop
- c.** Limaçon without an inner loop
- d.** Cardioid

54 Classify the curve $r = 3 + 5\sin \theta$.

- a.** Limaçon without an inner loop
- b.** Circle through the pole
- c.** Cardioid
- d.** Limaçon with an inner loop

55 Classify the curve $r = 7 + 5\cos \theta$.

- a.** Circle through the pole
- b.** Limaçon without an inner loop
- c.** Limaçon with an inner loop
- d.** Cardioid

- 56 Classify the curve $r = 5 + 5\sin \theta$.
- a. Limaçon without an inner loop
 - b. Cardioid
 - c. Circle through the pole
 - d. Limaçon with an inner loop
- 57 Find the smallest positive θ where the rose $r = 3\sin(4\theta)$ passes through the pole.
- a. $\frac{\pi}{4}$
 - b. $\frac{\pi}{16}$
 - c. $\frac{3\pi}{4}$
 - d. $\frac{\pi}{8}$
- 58 Find the smallest positive θ where the rose $r = 4\cos(6\theta)$ passes through the pole.
- a. $\frac{7\pi}{12}$
 - b. $\frac{\pi}{24}$
 - c. $\frac{\pi}{6}$
 - d. $\frac{\pi}{12}$
- 59 Find the smallest positive θ where the rose $r = 4\sin(4\theta)$ passes through the pole.
- a. $\frac{\pi}{16}$
 - b. $\frac{\pi}{4}$
 - c. $\frac{\pi}{8}$
 - d. $\frac{3\pi}{4}$

- 60 Find the smallest positive θ where the rose $r = 9\cos(4\theta)$ passes through the pole.
- a. $\frac{\pi}{16}$
 - b. $\frac{\pi}{4}$
 - c. $\frac{5\pi}{8}$
 - d. $\frac{\pi}{8}$
- 61 Find the smallest positive θ where the rose $r = 8\cos(5\theta)$ passes through the pole.
- a. $\frac{3\pi}{5}$
 - b. $\frac{\pi}{10}$
 - c. $\frac{\pi}{5}$
 - d. $\frac{\pi}{20}$
- 62 The limaçon $r = 7 + 14\cos \theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.
- a. $\frac{2\pi}{3}$
 - b. $\frac{\pi}{2}$
 - c. $\frac{5\pi}{6}$
 - d. $\frac{4\pi}{3}$
- 63 The limaçon $r = 7 + 14\sin \theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.
- a. $\frac{7\pi}{6}$
 - b. $\frac{11\pi}{6}$
 - c. $\frac{5\pi}{6}$
 - d. $\frac{\pi}{2}$

64 The limaçon $r = 3 + 6\cos \theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{2\pi}{3}$

b. $\frac{5\pi}{6}$

c. $\frac{\pi}{2}$

d. $\frac{4\pi}{3}$

65 The limaçon $r = 8 + 16\sin \theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{5\pi}{6}$

b. $\frac{\pi}{2}$

c. $\frac{7\pi}{6}$

d. $\frac{11\pi}{6}$

66 The limaçon $r = 1 + 2\cos \theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{2\pi}{3}$

b. $\frac{\pi}{2}$

c. $\frac{4\pi}{3}$

d. $\frac{5\pi}{6}$